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Abstract: In modern manufacturing, there is an increasing demand for reliable in-process measurement methods directly on large CNC machine tools, eliminating the need to transport workpieces to metrological laboratories. This study assesses the capability and applicability of an articulated arm coordinate measuring machine and a machine tool touch-trigger probe when measuring to a specified tolerance of 0.05 mm in a production environment. Experiments were conducted using the KOBA calibration standard and included measurements with and without applying the articulated arm coordinate measuring machine leapfrog method. The results were evaluated according to ISO 22514-7:2021 and ISO 14253-1:2017, which establish criteria for measurement system capability. The findings revealed that neither measurement system met the capability requirements of ISO 22514-7:2021, particularly due to unsatisfactory Q_{MS} and C_{MS} values. However, under ISO 14253-1:2017, both systems were deemed conditionally suitable for verifying conformity to the specifications, with the articulated arm coordinate measuring machine showing lower applicability when using the leapfrog method. This research supports the idea that unreasonable demands for compliance with current standards may lead to questioning of the systems that previously met older standards. The study contributes to the ongoing discussion on integrating advanced metrological tools into the manufacturing process and underscores the need for careful evaluation to ensure the capability and reliability of measurement systems in industrial practice.

Keywords: measurement system capability; articulated arm coordinate measuring machine; touch-trigger probe; on-machine measurement; on-site measurement; large scale measurement



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1. Introduction

In the global machining industry, there are stringent demands for accuracy and efficiency in manufacturing. Machined parts are often geometrically complex, have low rigidity, tend to deform, or are made from materials that are difficult to machine [1]. These factors can make it challenging to complete the machining process in a single step. During production, situations arise where parts need to be reworked to meet specified tolerances. Additionally, it is sometimes necessary to machine the parts with very small machining allowances, such as castings, forgings, or 3D-printed blanks. In these cases, it is crucial to know the size and distribution of the allowance before machining. Accurate in-process dimensional measurement is essential for precise machining in these scenarios. This can be achieved using Coordinate Measuring Machines (CMMs), conventional hand-held measuring tools, or directly on the machine tool itself (on-machine measurement) [2].

Stationary CMMs have been a key technology in precision engineering for decades, but their use presents several challenges [1,3] as follows:

1. Most CMMs require installation in a laboratory to ensure stable metrological conditions (temperature, vibrations, etc.). The metrological requirements for CMMs remain high due to their high accuracy.
2. Measuring on a CMM presents logistical challenges associated with transporting parts to the metrology laboratory. While this may not be a major issue for smaller parts, the logistics and costs associated with transporting oversized components are critical.
3. If a workpiece is found to be out of tolerance, it must be returned to the production area for rework and re-clamped on the machine tool, introducing potential errors into the manufacturing process.
4. These factors can lead to delays in the production schedule.

When using conventional hand-held measuring tools, we are often limited to measuring simple features such as distances, hole diameters, or runouts. However, inspecting certain workpieces may require the design and manufacture of special fixtures, which is time-consuming and uneconomical. Hand-held measurement is also highly dependent on the operator's experience and is difficult to apply to oversized workpieces.

Automated inspection of workpieces directly on manufacturing machines offers a solution to effectively increase not only production productivity but also process reliability. The high demand from machine tool users drives the continuous development of new methods for measuring workpieces on machine tools.

On-machine measurement (OMM) is defined as a measurement process where the object being measured is within the working area of the machine tool and the measuring equipment is also within the working area or at least in close proximity [2]. A typical method for measuring workpieces directly on the machine tool is the use of an integrated touch-trigger probe (TTP). This method of measurement offers several advantages:

1. The main advantage of OMM is that it eliminates the need to transport the workpiece to a metrology laboratory, thus avoiding the problems with re-clamping the workpiece for rework.
2. Measuring directly on the machine saves significant time and ensures adherence to the production schedule.
3. Under the right measurement conditions, the costs associated with using a touch probe are lower than those associated with the rejection of non-compliant products, leading to increased profitability, product quality, and customer satisfaction.
4. Modern machine tools are manufactured with very high accuracy, and with proper monitoring, maintenance, and calibration, they can maintain these tolerances, making them suitable for use as measurement systems (MSs).
5. The key advantage of using a touch probe is its automation, which reduces the need for human intervention [1,2].

Despite these advantages, there are still doubts as to whether it is possible to ensure the appropriate metrological conditions and rely on a production machine as a measuring device [4]. Besides concerns about the measurement environment, the complexity of measurement programming, and the interruption of productive machine time, there is also a general mistrust of using the same tool for machining and measurement, which could result in transferring the machining errors to the measurement results. Consequently, the measurement process is not independent of the manufacturing process [5]. These specific challenges of OMM can be mitigated by modern technologies and strategies, such as advanced calibrations, the use of temperature-compensated measuring tools, or the integration of systems for automatic monitoring and correction [6,7].

Some of the challenges associated with the use of stationary CMMs and touch probes can be addressed by utilizing portable coordinate measuring systems. Such systems can be transported directly to the part being measured, solving the logistical issues of transporting parts to the laboratory for measurement and allowing the measurement of parts while they are still clamped on the machine tool. The programming of portable CMMs is generally more flexible, and they are designed to operate in production environments. These systems include articulated arm coordinate measuring machines and laser trackers, which are fully

capable CMMs and can be used anywhere within a manufacturing facility, at any stage of production. If required by the application, these systems can also perform final inspections in a controlled metrological environment. The downside of these measuring devices in the context of current trends is their limited potential for automation.

All these measurement methods shall meet the metrological requirements for the specific measurement task. The applicability of an MS (including an OMM system) can be evaluated by verifying conformity with the specifications according to the ISO GPS system or by methods demonstrating MS capability, such as ISO 22514-7:2021, MSA:2010, or VDA 5:2011 [8].

This article focuses on the applications of OMM for in-process inspection, eliminating the need to transport workpieces to a metrology laboratory, and on the capability of MSs that make this possible. Conventional hand-held measurement methods are not addressed. While hand-held measurement can be very accurate, it cannot be considered highly efficient in terms of current demands for autonomous, intelligent manufacturing. Both integrated touch probe measurement and portable measuring systems are computer-supported, enabling the collection and analysis of large volumes of data for continuous improvement.

A case study is presented to illustrate this capability assessment. The entire process was carried out on an HCW 3 horizontal milling and boring machine using an RMP60 touch probe and a Hexagon Absolute Arm 8725-6-8084-FA. The experiments and results show that neither MS fully meets the capability requirements according to ISO 22514-7:2021, but both systems demonstrate conditional applicability for verifying conformity to specifications according to ISO 14253-1:2017 under specific conditions. The main aim of this study is to assess whether articulated arm coordinate measuring machines (AACMMs) and touch-trigger probes (TTPs) can be successfully integrated into the in-process control process directly on CNC machines, thereby enabling more efficient production. The findings support the idea that excessive demands for compliance with the current standard may lead to the improper questioning of MSs that previously met older standards. This article contributes to the ongoing discussion on the use of advanced metrological tools in the manufacturing process and highlights the need for careful evaluation of MS capability in industrial practice.

2. State of the Art

In the field of workpiece measurement, technologies utilizing CNC machining centers and coordinate measuring systems have seen significant advancements. These developments address the need for integrated measurement within the production process, thereby increasing efficiency and reducing the time required for quality control. This chapter provides an overview of the current state of these technologies, focusing on touch-trigger probes (TTPs) and articulated arm coordinate measuring machines (AACMMs). The overview includes relevant research, various practical applications, and key challenges associated with these technologies.

2.1. State of the Art of Workpiece Measurement with Touch-Trigger Probes

To date, numerous studies have focused on the possibility of using CNC machining centers as measuring devices, with integrated TTPs serving as the key tool. The integration of measurement processes directly into the machining equipment offers the potential to significantly reduce non-productive times by eliminating the need to transfer workpieces to specialized measuring devices, such as CMMs. The accuracy of measurements using the TTPs on CNC machine tools is influenced by several factors, including the geometric accuracy of the machine, thermal stability, and the machine's ability to compensate for errors.

Zhuang et al. [1] provide a comprehensive overview of the latest research, technologies, and applications of OMM using TTPs. OMM offers several advantages over traditional methods, such as CMMs, including lower costs, ease of use, and reduced environmental

and operator requirements. As a result, it is becoming an increasingly popular solution for dimensional control in the production of complex components. Research in this field focuses on improving measurement accuracy and efficiency, as well as exploring various OMM applications. Key topics include metrology process planning, process optimization, and the evaluation of measurement capability and traceability. Similar to other publications, considerable attention is paid to the challenges associated with OMM. One concern, for example, is that this method detracts from the actual machining time. However, this argument neglects to consider the time-consuming nature of manual measurements. Another challenge is gaining trust in OMM results, which can be addressed through advances in probe design, compensation methods, and process optimization. The article also provides a detailed overview of the main technologies necessary for successful OMM, such as CNC machining centers, TTPs, and specialized software. TTPs are designed for repeatable, task-specific measurements, while the software is used for calibration and processing of the measured data. Furthermore, the article highlights future research directions, including the integration of OMM into engineering processes to improve accuracy and efficiency, as well as the need to develop comprehensive software solutions. Overall, the article demonstrates opportunities for future research, which is crucial for improving manufacturing processes across a wide range of industries.

Krawczyk et al. [9] explore the role of OMM in enhancing the sustainability of machining processes. It highlights how integrating measurement directly into CNC machine tools reduces manual interventions, improves precision, and minimizes waste and energy consumption. The article compares two types of TTPs—kinematic resistive and strain gauge—and evaluates their performance in real production conditions. The authors emphasize the environmental and economic benefits of OMM, such as reduced downtime, better resource utilization, and safer working conditions. The findings underline OMM's critical role in sustainable and efficient manufacturing, especially in high-precision industries like aerospace.

Among the specific articles assessing the capability of machine tools as measuring devices, the following are particularly noteworthy. Takaya [10] reviews in-process and on-machine measurement techniques for improving machining accuracy and managing process and product quality. It discusses advancements in metrology for closed-loop control systems and their application to ultra-precision machining, highlighting the integration of machine tools, measuring instruments, and control strategies. Key topics include measurement uncertainty, process capability, and the role of these techniques in addressing geometric errors, surface quality, and tool wear. The study emphasizes the growing importance of holistic measurement approaches in achieving high accuracy and efficiency in modern manufacturing processes, particularly in ultra-precision and complex machining environments.

Holub et al. [4] examined the application of the MS capability indices C_g and C_{gk} for evaluating the capability of measurements performed using TTPs on CNC machine tools. [4]. The index of measuring device capability C_g (capability index) evaluates the precision (repeatability) of the measuring device. The extended capability index C_{gk} evaluates both the precision and accuracy of the measuring device. These indices help quantify how reliably a machine tool can measure within specified tolerances. In this case, the tolerance was set at $T = 0.015$ mm on the Y-axis of the CNC machine tool. The article proposed a method for eliminating systematic errors resulting from the machine's geometric accuracy. Following the application of the correction, the machine tool was evaluated as being capable of meeting the specified tolerance. It was found that, with appropriate calibration and under controlled environmental conditions, CNC machines can effectively fulfill the dual role of both production and measurement devices.

Szyszkla et al. [5] evaluate the performance of various TTP configurations through MSA methods, particularly focusing on Repeatability and Reproducibility (R&R) analysis. Conducted under real shop floor conditions in the aerospace industry, the study aimed to identify the measurement accuracy and limitations of TTP systems for closed-loop

machining of large jet engine cases. Key findings include R&R values below 15% in most scenarios, highlighting acceptable system performance. The study emphasizes the role of adaptive measurement systems in achieving precision for complex geometries and addresses challenges such as thermal variability, probe design, and calibration issues.

Ding et al. [11] investigated the potential of laser sensors to enhance OMM capabilities on CNC machine tools. An iterative calibration method for the laser sensor was introduced, and the findings showed that after five iterations of calibration, the difference between the calibrated and actual center of the calibration sphere was reduced to 4.7 μm , which was considered acceptable. A comparison with the TTP revealed that while the TTP was more accurate, the laser sensor was more efficient, with uncertainties ranging between 4.8 and 5.8 μm for the laser sensor and 2.4 and 3.7 μm for the touch probe. The MS capability was evaluated in accordance with ISO 22514-7:2021. The tolerance ranges in which the probes were deemed capable were calculated as 16.0–24.7 μm for the TTP and 30–38 μm for the laser sensor.

Blecha et al. [12] raise questions about whether CNC machine tools equipped with TTPs can achieve the required level of accuracy for applications, particularly in the production of high-value workpieces. It was found that under controlled conditions, where stable temperature and carefully set machine parameters were maintained, CNC machines achieved measurements with minimal deviation. Exact deviation values were not specified, but the importance of meticulous machine setup and calibration was emphasized, as changes in machine compensation and geometric accuracy settings directly impacted the measurable tolerance and precision. The study concluded that optimizing these settings could lead to significant improvements in measurement accuracy, which is crucial for the effective use of CNC machine tools as measuring devices. A key focus of this research was to introduce the capability assessment of measuring systems (MSs) on CNC machine tools with integrated TTPs, based on VDA 5 and ISO 14253-1:2017 standards. The tests determined the minimum allowable tolerance of the assessed workpiece that could be measured on the CNC machine tool.

A key theoretical basis for this case study is the recent article by Blecha et al. [8]. This article explored the possibilities of using CNC machine tools as measuring systems for dimensional control. The essential rationale behind this research is the potential improvement in manufacturing productivity through OMM, which saves time by eliminating the need to transfer the workpiece to a separate measuring device. However, to ensure that such MSs meet the required measurement standards, the reliability of the measured data must be ensured. The article presented a new approach to evaluating the capabilities of CNC machines as measuring systems, relying on ISO standards and statistical methods (e.g., ISO 22514-7:2021 and VDA 5:2011). This approach was tested on a three-axis vertical CNC machining center with a cross table, the MCV 754, and compared with the Hexagon GLOBAL SPEED CMM. The results showed that, at a tolerance of 0.015 mm, the CMM was able to meet the requirements according to ISO 22514-7:2021, while the CNC machine was not. However, below this tolerance, both machines were found applicable for verifying the conformity with technical specifications according to ISO 14253-1:2017. In the experimental section of the article, the length of the KOBA standard was measured in modes with and without geometric error compensation. The CNC machine achieved a minimum tolerance of 0.0262 mm with error compensation (ENC 1) and 0.0454 mm without compensation (ENC 0). In contrast, the CMM achieved a minimum tolerance of 0.0137 mm. The article emphasizes that an uncritical adherence to the requirements of ISO 22514-7:2021 could lead to the questioning of systems that have previously been proven effective under earlier standards. The results clearly show that, although CNC machines do not match the accuracy of CMMs, they can be used as measuring systems under certain conditions. In conclusion, the authors recommend that the requirements of the standard should be critically assessed when implementing CNC machine tools as measuring systems in order to minimize costs and enhance production efficiency.

2.2. State of the Art of Workpiece Measurement with AACMM

AACMMs are portable coordinate measurement machines. The core advantage of using AACMMs is that the entire MS, including accessories, can be packed into a single transport case and taken to any location for the measurement application. It is a highly flexible MS that provides real-time feedback on the dimensions and geometry of parts directly at the production site. AACMMs are available in various sizes and are generally suitable for rapid inspection of parts ranging between 2000 and 3000 mm, and for geometrically complex parts around 1500 mm. In many cases, AACMMs can also be used for large-scale metrology tasks, as their measurement range can be extended using accessories such as the Leapfrog Kit or GridLock [13]. The “Move Device Position”, or leapfrog, is a technique that extends the measurement volume of portable measuring devices. A set of reference points is measured from the first position using the portable measuring system, after which the system is moved to a new position, and the reference points are re-measured. This set of reference points links the new position of the MS to the original coordinate system, enabling measurements and evaluations to be conducted in a unified coordinate system [14].

AACMMs offer the advantage of being used directly at the production site of the component, though their accuracy is inherently lower than that of stationary CMMs of similar size [15]. To date, many articles have been published focusing on improving the measurement accuracy of AACMMs. A significant proportion of these studies has primarily addressed the factors influencing measurement uncertainty when using AACMMs, as well as the possibilities for improving this uncertainty [16,17].

In general, there are two ways to reduce the uncertainty of AACMMs. The first is to improve the manufacturing and assembly accuracy of the arms. Here, the only option is to enhance the accuracy of individual components (encoders, materials, arms, sensors, etc.), but the guarantee of reducing the uncertainty of the entire measuring device is limited. This method also increases the production costs of AACMMs. The second approach is kinematic calibration, which is based on the theory of robot parameter calibration. Correctly identified and calibrated parameters are then used to compensate for AACMM errors [17,18]. Articles [19–21] utilize mathematical models and methods to accurately identify key AACMM parameters. Kinematic modeling based on the Denavit–Hartenberg method and analysis using the Jacobian matrix help identify errors and relationships between individual parameters. Separating the kinematic parameters allows for higher accuracy, leading to more efficient and reliable measurements.

Many articles share common information and conclusions regarding AACMM accuracy. In most studies, the Denavit–Hartenberg method is used for kinematic modeling of AACMMs. This method provides an accurate description of kinematics and helps simulate measurement uncertainties. Another common element is the use of Monte Carlo simulation, which allows for the quantification of measurement uncertainties and for the variability of individual factors to be considered. These methods enable a more accurate estimation of measurement uncertainties in industrial conditions and improve the overall reliability of measurements using AACMMs.

González-Madruga et al. [22,23] confirm that the human factor can have a significant impact on the measurement accuracy in AACMMs. Although calibration methods often account for the machine itself, the operator’s influence on measurement errors is not always adequately considered. Therefore, the methodologies presented in these articles emphasize the necessity of operator qualification and the verification of measuring systems (MSs) as integral components of the measurement process. Similar to previous studies, the use of the Denavit–Hartenberg method and Monte Carlo simulations for estimating the measurement uncertainties is also mentioned.

From the conducted research, it can generally be concluded that most experiments with AACMMs were performed in controlled laboratory environments. This allows for greater control over the measurement conditions and eliminates some variables that could affect the results. However, in the case of AACMMs, these conditions do not fully reflect reality. Only a few of the available publications were found to have been conducted in a

production environment, or at least to have simulated such an environment. Stepien [24] proposed compensation methods for issues such as vibrations and temperature fluctuations affecting cylindricity measurement in a production environment, emphasizing proper calibration. Acero et al. [25] introduce a method for verifying the accuracy of AACMMs using a laser tracker and an indexed metrological platform in a production environment, allowing for measurement error compensation. El Asmai et al. [26] describe a method to quickly and reliably evaluate AACMM performance directly in the production environment by means of comparative measurement on reference objects. Nagao et al. [27] outline the calibration of machines with a parallel mechanism using AACMMs. The experiments were conducted in a production environment to ensure that the calibration conditions closely resembled real-world scenarios. Saito et al. [28] propose a new kinematic model with a “map of errors” for the rotational axes. The experiments were conducted on a CNC machine tool in an environment simulating real production. Wang et al. [17] describe a method for calibrating AACMMs directly in a production environment. The method uses a high-precision machine tool as a calibration device, with the AACMM mounted on the worktable and the calibration artifact attached to the spindle. The machine tool enabled the creation of a “virtual calibration artifact” with calibration points calculated using a Hammersley sequence. The method was validated by the tests according to ASME B89.42-2004 and ISO 10360-12 standards. Measurement uncertainty was significantly reduced after calibration. This approach allows for calibration directly in the production without additional equipment, increasing the efficiency of the calibration process.

The results regarding the determination of the capability of AACMMs are quite limited. Gaska et al. [29] present a method that suggests the possibilities for evaluating the measurement ability of AACMMs for specific measurement tasks. This method is designed for quick and specific evaluation of AACMM measurement ability. It combines standardized measurement procedures with reference values that are key to determining whether the device can perform measurement tasks with sufficient accuracy. A crucial element is the calibration block, which was measured in three different positions and configurations of the AACMM to identify deviations from nominal values when measuring length, flatness, perpendicularity, and parallelism. The measured values are compared with predefined tolerances for each parameter. These tolerances are set either based on previous measurements or according to industry standards. The tolerance was set at 0.08 mm for lengths, 0.01 mm for flatness, 0.05 mm for perpendicularity, and 0.04 mm for parallelism. If the measured values remain within these limits, the device is considered capable of performing the specific measurement operation. From the specific results, it is notable that the highest measured length deviation reached 0.0756 mm, the deviation in flatness measurement was 0.0074 mm, the highest deviation in perpendicularity was 0.0465 mm, and the highest parallelism deviation was 0.0364 mm. Although this is not a specific evaluation of MS capability, this method allows users to efficiently determine the AACMM’s measurement capability for various measurement tasks.

3. Research Approach

The main focus of this case study is the assessment of the capability of an AACMM and an integrated workpiece touch-trigger probe for measurement on a CNC machine tool in a production environment. The research is aimed at evaluating the capability of these systems and their effectiveness in in-process control. This study assumes that the use of AACMMs and TTPs on CNC machine tools will achieve sufficient measurement capability, comparable to traditional methods such as stationary CMMs. Additionally, it is expected to maintain higher efficiency and accuracy without the need to transport workpieces to a metrology laboratory. Furthermore, it is assumed that the use of a TTP and a portable measuring system directly on the machine tool can lead to time savings and cost reductions associated with the logistics of transporting workpieces. The application is based on methods proposed in previous research, FW01010012 Research and Development in the Field of Improving the Quality of Production of Large Workpieces—Precise and Repeatable

Setup and Measurement of Workpieces, where the authors developed a practical proposal of leapfrog measurement methodology and an assessment of measuring device capability [30]. This methodology outlines specific procedures for measuring and evaluating the capability of MSs in a production environment. Placing the research in a production environment offers several advantages but also presents challenges. Measuring directly in production allows for rapid in-process control and immediate response to deviations, which can significantly increase production efficiency. On the other hand, the production environment is often exposed to factors such as vibrations, temperature changes, and dust, which can negatively affect measurement accuracy. Another challenge is the need for skilled personnel who can correctly set up and operate the measuring equipment in demanding production conditions. These factors can increase the variability in the measured data and make it difficult to achieve consistent results. Therefore, it is important to consider these conditions when evaluating the capability of measuring systems and implementing measurement methods. The experiments were carried out on the HCW 3 horizontal milling and boring machine (SMT, CZ). It is a large CNC machining center. The workpieces weigh up to 20 tons and have dimensions of $7 \times 3 \times 2$ m, with tolerances of the controlled dimensions ranging from 0.07 mm to 0.5 mm. Production and measurement on this machine represent very current demands from companies capable of machining oversized components, where the transportation of products is a significant issue. In this group of manufacturers, there is a strong demand to perform the measurements directly on the machine tool and to ensure that these measurements are reliable for their customers. This case study builds on the article by Blecha et al. [8], which emphasizes that uncritical adherence to the current ISO 22514-7:2021 or VDA 5:2011 standards may call into question MSs that have proven effective under previous standards. The results show that, although CNC machines do not achieve the precision of CMMs, they can be used under certain conditions as measuring systems for verifying conformity with technical specifications. The aim of this case study is to take into account the new approach to capability and applicability evaluation presented in the article by Blecha et al. [8], which is based on the evaluation of MS capability according to ISO 22514-7:2021 and the evaluation of MS applicability for verifying the conformity with technical specifications according to ISO 14253-1:2017, to replicate the evaluation procedures, and to verify these methods under conditions more representative of a real production environment. The setup of the AACMM and TTP on CNC machine tools was performed according to the specified measurement procedures. The measurements were conducted on a KOBA calibration standard with ceramic gauges. Each calibrated length was measured both with and without the leapfrog method using the AACMM. Leapfrog measurement with the AACMM is particularly important when measuring oversized parts, so it is necessary to verify the AACMM's capability when using leapfrog. The evaluation included the calculation of combined measurement uncertainty and the performance ratio (Q_{MS}). In this research, the reference tolerance was set at 0.05 mm. This value was chosen to represent current industrial requirements in the production of precise oversized components. The research also focused on investigating the minimum achievable measurement tolerance. The results of this study contribute to a better understanding of the capability of measuring systems and their implementation in industrial practice. The research also supports a critical evaluation of the requirements for standards to ensure their suitability for modern production environments, thereby achieving higher efficiency and lower costs.

4. Materials and Methods

4.1. Equipments Used

For the experimental measurements, a Renishaw RMP60 workpiece TTP with radio signal transmission was used, which is suitable for medium and large machining centers. To demonstrate measurement with the AACMM, the Hexagon Absolute Arm 8725-6-8084-FA model was employed. The experiments with the AACMM were conducted using a 15 mm diameter touch probe with a metal tip, and the measurement program was written in the

PC-DMIS software (2020 R2 SP14). This software is supplied with the AACMM and allows for precise measurements and evaluation of the results. The measurements were carried out on the HCW 3 horizontal milling and boring machine, which was produced by SMT a.s. The SMT a.s. is one of the leading world producers of horizontal boring and milling machines, horizontal lathes, rotary tables, and special accessories. The HCW 3 machine enables precise and efficient machining of oversized and geometrically complex products, as detailed in Table 1. Figure 1 shows the specific HCW 3 machine in the production environment. More information about the machine can be found in reference [7].

Table 1. Parameters of horizontal boring and milling machine HCW 3-200 NC.

Parameter	Value
Spindle Diameter (mm)	200
Main Motor Power (kW)	129
Max. Spindle Speed (rpm)	2000
Ram extension Z (mm)	1600
Spindle Extension W (mm)	1400
Max Headstock Traverse Y (mm)	9000
Max. Traverse X (mm)	30,000



Figure 1. The HCW 3 horizontal milling and boring machine.

The Calibration Standard with Ceramic Castellations KOBA (hereinafter referred to as KOBA) was used as the reference standard for measurement. Table 2 presents the nominal lengths and the corresponding actual lengths from the calibration performed. The measurement uncertainty for the center-to-center distance with a coverage factor $k = 2$ is as follows:

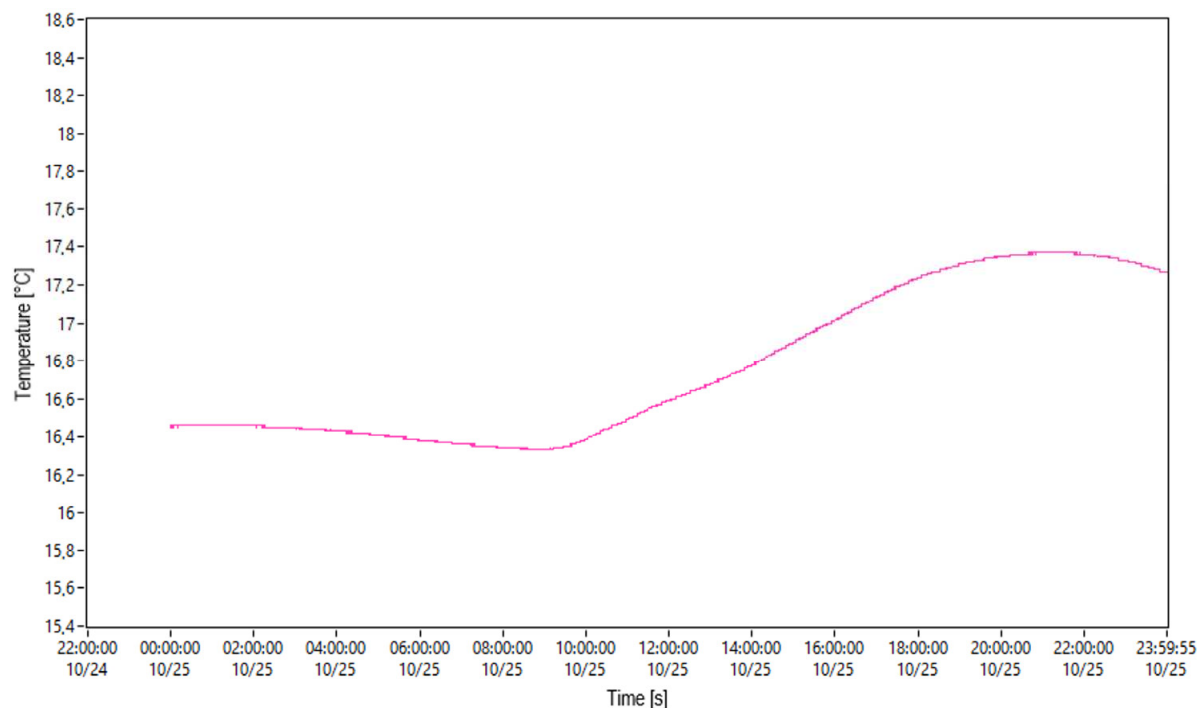
$$U_{(k=2)} = \left(0.30[\mu\text{m}] + 0.8 * 10^{-6} * L[\text{m}] \right) [\mu\text{m}]. \quad (1)$$

Table 2. KOBA step gauge nominal and measured value.

Length Number	Nominal Value [mm]	Distance from Measuring Surface 0 [mm]
0	0	0.00000
1	41.3	41.30148
2	281.2	281.19792
3	481.1	481.09800

4.2. Methodology

In order to verify the capability evaluation possibilities, demonstration experiments were conducted in an industrial environment, where the production specialty is large and heavy workpieces, steel structures, as well as light and medium-weight weldments. The environment in the production hall during the tests was relatively stable, both due to the season when temperature fluctuations are not extreme, and the surrounding production processes. The total measurement time for the calibration standard using both measuring systems was approximately 3 h. Figure 2 shows the daily temperature profile on the Y-axis at the height of the KOBA standards measurement obtained from the HCW 3 machine monitoring. The graph indicates that the difference between the lowest and highest temperatures during the day is approximately 1 °C. This range can be considered acceptable given the real production environment.

**Figure 2.** Daily temperature profile during the measurement—25 October 2023.

Before the measurement process began, the calibration standard was prepared, which primarily involved leveling the standard on the worktable of the HCW 3 machine to a tolerance of ± 0.01 mm using a 125 mm Johansson gauge block and a Digico 400 digital dial gauge. The correct interpretation of the measured error signs was crucial during leveling in order to minimize the overall error. The KOBA standard consists of five calibrated lengths, of which three were measured: the shortest possible distance of 41.3 mm, the medium distance of 281.2 mm, and the longest distance of 481.1 mm (see Figure 3).

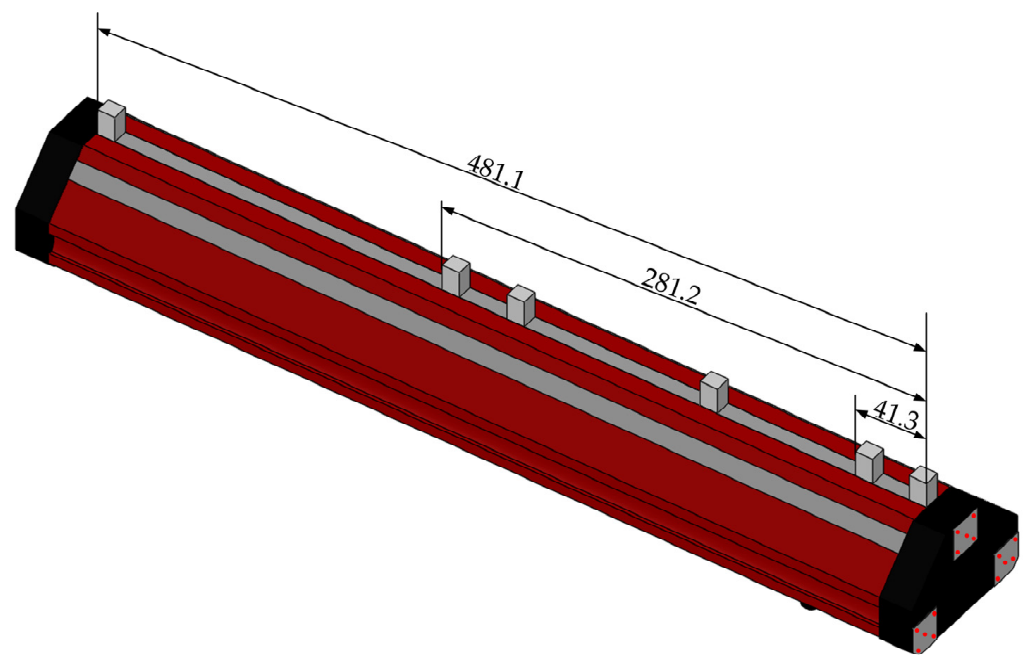


Figure 3. Illustration of the measured lengths and the positions of points for leveling the KOBA standard.

The AACMM was mounted on the extended spindle of the HCW 3 machine. The preparation of the measurement strategy with the AACMM in the metrology software involved defining the coordinate system and selecting the touch probes and measured characteristics. Before starting the measurement, the touch probe was calibrated by measuring a reference sphere of known diameter, where nine points were captured, and the software calculated the diameter, minimum and maximum values, and standard deviation. The standard was further aligned using 15 points distributed across three frontal surfaces of the KOBA standard (see Figure 3). The measurement itself consisted of 15 repetitions on the three frontal surfaces of the KOBA standard. Each ceramic Johansson gauge block was measured with a single point approximately at its center. During the measurement, emphasis was placed on the stability of the standard's temperature and the avoidance of the influence of external forces. When performing the measurements with the arm, the so-called leapfrog method was also used, which is commonly applied in practice to measure the parts larger than the AACMM's measurement volume. Leapfrog is a measurement method that allows the AACMM to be moved within the workspace while maintaining a consistent coordinate system. The leapfrog was executed by measuring three sets of points, where the fixtures with an inverted cone base (points 1–3 in Figure 4) represented the working space. The arrangement of the fixtures must not form an equilateral triangle. The fixtures were measured using a probe with a diameter larger than the cone base to prevent the probe from moving. After measurement, the arm was shifted by moving the table a defined distance, taking into account the placement of the leapfrog fixtures and maximizing the available space. The fixtures were then remeasured in the same order. The metrology software calculated the residual error in the X, Y, and Z axes, as well as the spatial error, which should not exceed the repeatability error of the arm (see Figure 4) [14].

When measuring with the RMP 60 workpiece TTP, the same setup of the KOBA calibration standard was used as with the AACMM measurement. The TTP was mounted in the spindle (see setup in Figure 5), and the probe tip diameter was calibrated. Using a pre-programmed measurement cycle, the defined distances were measured, with 15 repetitions performed.



Figure 4. Measurement setup with the Hexagon Absolute Arm 8725, including the indication of Leapfrog Kit position.



Figure 5. Measurement setup with touch-trigger probe RMP60.

The evaluation of the measurement results was primarily conducted according to the standards ISO 22514-7:2021 and ISO 14253-1:2017, in accordance with the methodology of Blecha et al. [8,31,32].

4.3. Theoretical Basis for Measurement System Capability and Applicability Assessment

The evaluation of MS and process capability in engineering is crucial for ensuring production accuracy and quality. One of the key methods is Measurement System Analysis (MSA), which focuses on the analysis and evaluation of MS variability. MSA includes repeatability and reproducibility (R&R) studies, which analyze the variation caused by measurement instruments and operators, without explicitly taking measurement uncertainty into account. VDA 5 and ISO 22514-7:2021 share common elements, particularly regarding statistical methods for evaluating the capability of MSs and processes. Both

methods use similar calculations and approaches to determine measurement uncertainty and capability, which explains their interrelation. VDA 5, developed specifically for the automotive industry, provides guidelines for the systematic evaluation of MS capability. It focuses on measurement accuracy and includes methods for analyzing measurement uncertainty. VDA 5 contains procedures for quantifying the uncertainty and the capability of MSs, which is crucial for ensuring high-quality and reliable results in the highly regulated automotive industry. ISO 22514-7:2021 focuses on statistical methods for evaluating process capability and performance. This standard provides a framework for the use of various statistical techniques to improve and maintain process quality, including capability analysis. ISO 22514-7:2021 also incorporates measurement uncertainty into capability assessments, allowing for a comprehensive view of the performance of MSs and processes. Both methods utilize similar statistical tools, such as control charts, process capability analysis (C_p , C_{pk}), and analysis of variance (ANOVA). These tools are used to evaluate the variability and capability of MSs and processes. Both VDA 5 and ISO 22514-7:2021 include procedures for quantifying and analyzing measurement uncertainty. This ensures that capability evaluation is not just about measuring the average values, but also about understanding and managing the variability and uncertainty that can affect results. Both methods provide a systematic approach to evaluating MSs and processes, including defining criteria, collecting data, analyzing results, and implementing corrective actions based on the findings. ISO 14253-1:2017 primarily focuses on verifying the conformity with technical specifications and details procedures for incorporating the measurement uncertainty into this process. While not primarily intended for assessing MS capability, its principles and methods can be effectively applied for applicability evaluations by providing clear guidelines for incorporating measurement uncertainty. This ensures that MSs can accurately and reliably measure the required parameters, contributing to the overall capability assessment. It is important to note that while MSA focuses primarily on repeatability and reproducibility, standards such as VDA 5, ISO 22514-7:2021, and ISO 14253-1:2017 include measurement uncertainty, providing a more comprehensive and accurate view of the capability and applicability of MSs. The chronological development of these methods and standards reflects the increasing demands for measurement accuracy and reliability, corresponding to the needs of modern industry for detailed and systematic evaluation of MS and process capability. The capability evaluation procedures for the assessed MSs presented in the following chapters are consistent with the articles by Blecha et al. [8,12] and the relevant standards [31–33].

4.3.1. Procedure for Evaluating MS Capability According to ISO 22514-7:2021

The ISO 22514-7:2021 standard defines the procedure for evaluating the capability of measurement systems (MSs) and measurement processes (MPs) based on the theory of combined measurement uncertainty and clarifies whether a given measurement process meets the requirements for a specific measurement task.

Resolution RE is a key factor that affects the system's ability to detect small changes in measured values. The standard specifies that the resolution of the MS should be less than 5% of the tolerance (TOL), ensuring accurate detection of deviations within the measured tolerance. If the resolution is insufficient, it can negatively impact the overall capability of the system. In such cases, the capability evaluation should be suspended, as inadequate resolution calls into question the reliability of further calculations, such as the Q_{MS} and C_{MS} indicators.

$$\%RE = \frac{RE}{TOL}(100\%) \leq 5\%TOL. \quad (2)$$

The MS performance ratio (Q_{MS}) indicator evaluates the ratio between the expanded measurement uncertainty (U_{MS}) and the tolerance size TOL . The Q_{MS} value should not exceed 15%, i.e., the expanded measurement uncertainty should not be more than 15% of the total tolerance.

$$Q_{MS} = \frac{2U_{MS}}{TOL}(100\%). \quad (3)$$

The MS capability index (C_{MS}) monitors how many combined measurement uncertainties can fit within the given tolerance. The standard requires the C_{MS} to be at least 1.33, i.e., that at least 5.32 combined measurement uncertainties must fit within one-fifth of the tolerance.

$$C_{MS} = \frac{0.2(TOL)}{2 \cdot k \cdot u_{MS}}. \quad (4)$$

U_{MS} in Equation (4) represents the expanded measurement uncertainty and can be expressed by the equation:

$$U_{MS} = k(u_{MS}) \quad (5)$$

In Equations (4) and (5), u_{MS} is the combined standard uncertainty, and the expansion coefficient $k = 2$, as the approximately 95.45% coverage interval is used in accordance with the standard.

The relationship between the Q_{MS} and C_{MS} indicators is such that if Q_{MS} meets the required criterion (i.e., $Q_{MS} \leq 15\%$), the C_{MS} indicator will also always be compliant. This relationship stems from the fact that both indicators are based on the same parameters—specifically, the combined uncertainty of the MS (u_{MS}) and the size of the tolerance range (TOL). Both indicators share the same goal: to determine whether the MS is capable. Q_{MS} expresses the ratio of the expanded uncertainty to the tolerance range and ensures that the expanded uncertainty of the MS does not exceed 15% of the tolerance. C_{MS} , on the other hand, follows a similar goal but works with combined measurement uncertainty (u_{MS}) and focuses on how many of these uncertainties can fit within the given tolerance. Since both indicators are based on combined measurement uncertainty, if Q_{MS} meets the requirement, it means that the MS occupies less than 15% of the tolerance, which automatically guarantees that C_{MS} will also fall within the required range (greater than 1.33). Therefore, if the Q_{MS} indicator is within the required range, the C_{MS} indicator will also be compliant. This makes C_{MS} redundant in this context if Q_{MS} is used as the primary criterion for assessing capability. This may explain why some standards, such as VDA 5, do not include the C_{MS} indicator, considering Q_{MS} sufficient for evaluating the MS's capability. Figure 6 illustrates the individual steps of the MS capability evaluation process according to ISO 22514-7:2021. First, the resolution of the MS is verified, ensuring the ability to detect small deviations within the given tolerance. Next, the MS performance ratio is evaluated to determine whether the expanded measurement uncertainty, multiplied by two, occupies less than 15% of the tolerance. If both criteria are met, the test is successful. The diagram suggests that the capability indicator criterion is supplementary to this concept. The capability evaluation process in this article complies with the provisions of ISO 22514-7:2021; therefore, the C_{MS} capability index was evaluated in the standard manner.

The combined standard uncertainty of the MS u_{MS} (see Equations (4) and (5)) is determined using the following Equation (6)

$$u_{MS} = \sqrt{u_{cal}^2 + \max\{u_{EVR}^2, u_{RE}^2\} + u_{BI}^2 + u_{LIN}^2 + u_{MS_REST}^2}, \quad (6)$$

where the individual components from Equation (6) are calculated step by step according to the equations in Table 3.

The release of ISO 22514-7:2021 is stricter in several respects compared to the 2012 version, particularly in the area of MS capability evaluation. One of the key changes is the refinement of the requirements for the use of Maximum Permissible Error (MPE) values. While the previous version relied more on experimental methods for capability evaluation, the new version clearly states that the MPE value can be used, which simplifies calculations but also requires strict adherence to the procedure for correctly calculating the combined uncertainty. The calculation of combined uncertainty incorporating MPE provides a standardized way of assessing the system capability, allowing for quicker

implementation in the production process without the need for repeated experiments. The equation for calculating MPE is as follows:

$$u_{MS} = u_{MPE} = \frac{MPE}{\sqrt{3}}. \quad (7)$$

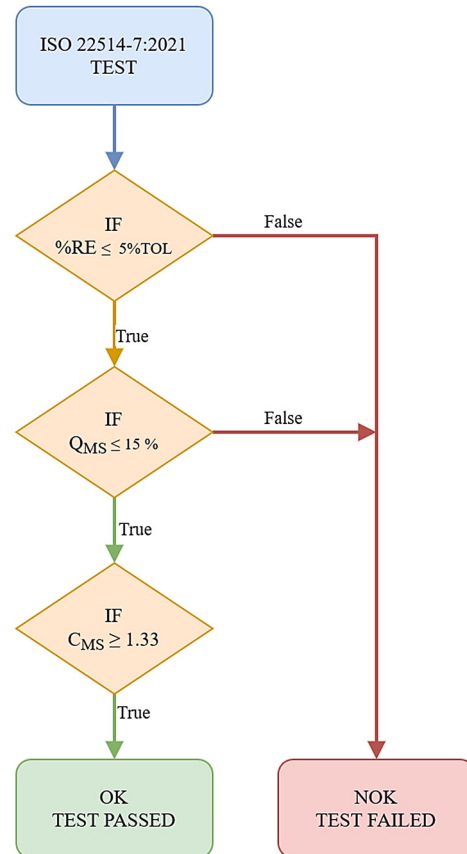


Figure 6. Flowchart of the MS capability assessment process according to ISO 22514-7:2021.

Table 3. Components of combined measurement uncertainty and their calculation.

Uncertainty Components	Symbol	Model
Uncertainty arising from resolution	u_{RE}	$u_{RE} = \frac{0.5 \cdot RE}{\sqrt{3}}$
Uncertainty arising from calibration	u_{CAL}	$u_{CAL} = \frac{U_{CAL}}{k_{CAL}}$
Uncertainty arising from bias	u_{BI}	$u_{BI} = \frac{ \bar{x}_g - x_m }{\sqrt{3}}$
Uncertainty arising from linearity	u_{LIN}	$u_{LIN} = \sqrt{MS_{LIN}}$
Uncertainty arising from the repeatability using reference standards	u_{EVR}	$u_{EVR} = \sqrt{\frac{1}{n-1} \sum_{i=0}^n (x_i - \bar{x}_g)^2}$ $u_{EVR} = \sqrt{MS_{EVR}}$
Other uncertainty components	u_{MS_REST}	If necessary, it is determined according to the instructions in GUM 1995.

If the MS is characterized by multiple MPE values, the calculation is carried out according to Equation (8).

$$u_{MS} = u_{MPE} = \sqrt{\frac{MPE_1^2}{3} + \frac{MPE_2^2}{3} + \dots} \quad (8)$$

Another tightening concerns the Q_{MS} indicator, where ISO 22514-7:2021 specifies that Q_{MS} shall not exceed 15%, i.e., that the combined measurement uncertainty shall not exceed 15% of the total tolerance, thereby increasing the requirements for the accuracy of MSs.

Along with this, the definition of the C_{MS} indicator has also been refined, with the standard requiring C_{MS} to be at least 1.33, which means that at least 5.32 combined uncertainties must fit within one-fifth of the tolerance range.

These stricter requirements put pressure on manufacturers to ensure that their MSs can meet these demands and potentially invest in more advanced and accurate measuring equipment that complies with the stringent requirements of ISO 22514-7:2021 [8].

If we designate the maximum allowable performance ratio for the MS as $Q_{MSmax} = 15\%$, we can also calculate the minimum tolerance TOL_{MIN} , for which, given the calculated expanded uncertainty U_{MS} , the specified condition (as well as $C_{MS} = 1.33$) is met:

$$TOL_{MIN} = \frac{2U_{MS}}{Q_{MSmax}}(100\%) = \frac{2U_{MS}}{15}(100\%). \quad (9)$$

4.3.2. Verifying the Conformity with the Specification According to ISO 14253-1:2017

According to ISO 14253-1:2017, conformity with the specification is verified when the measured value falls within the acceptance zone. The acceptance zone is the specification range reduced by guarding bands, taking into account the probability limit of compliance (see Figure 7).

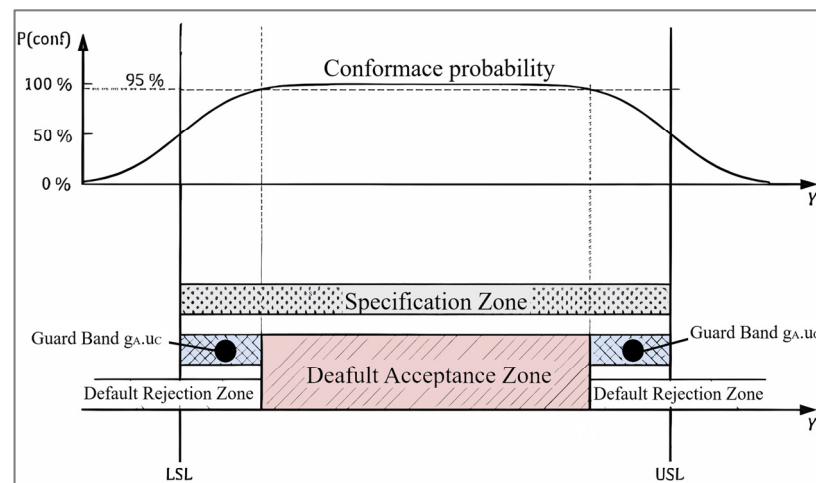


Figure 7. Illustration of relevant zones for verifying conformity according to ISO 14253-1:2017.

If the probability density of the measured values follows a normal distribution with a standard deviation significantly smaller than the specification range, then the standard probability limit of compliance of 95% corresponds to a guarding band factor value of $g_A = 1.65$, i.e., the width of the guarding band is equal to 1.65 times the combined standard uncertainty u_c . The width of the guarding band is given by the Equation (10):

$$\text{width of guarding band} = g_A u_c. \quad (10)$$

To determine the value of g_A , we calculate the ratio of tolerance TOL to the combined measurement uncertainty u_c using Equation (11):

$$RATIO = \frac{TOL}{u_c}. \quad (11)$$

The value of the guarding band factor g_A is then calculated so that the probability of conformity with the specification is exactly 95%. The maximum value of the guarding band factor is $g_A = 1.96$, which applies when the ratio (11) is exactly 3.92. Figure 8 shows the graph of the guarding band factor g_A values depending on the size of the $RATIO$ according to Equation (11).

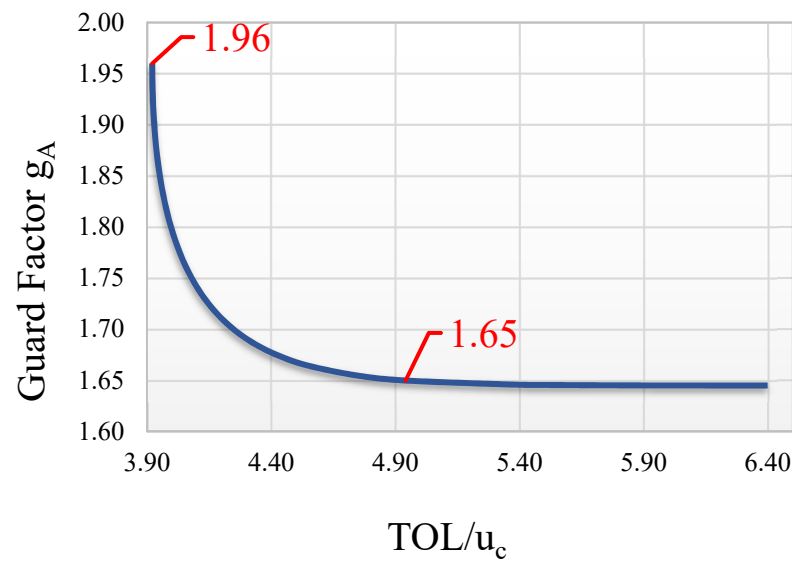


Figure 8. Graph of guarding band factor (g_A) values as a function of *RATIO* size.

Let us further assume that the value of the combined measurement uncertainty is estimated using the maximum permissible measurement error (MPE) according to the previous Equation (7).

$$u_{MS} = u_{MPE} = \frac{MPE}{\sqrt{3}}. \quad (12)$$

5. Results

5.1. The Results of Measurements Using the AACMM

Table 4 below presents the set of measurement results obtained with the AACMM in accordance with ISO 22514-7:2021. Based on the MS validation procedure, the values of three criteria were determined to assess whether the AACMM meets the measurement requirements for the specified tolerance of 0.05 mm. Of the obtained criteria, only the %RE resolution parameter meets the requirements, with a value of 2% *TOL* for both measurements without leapfrog and with leapfrog. The values of the other two criteria do not meet the requirements in either case; therefore, the MS is not capable of this tolerance value. Additionally, the minimum tolerance TOL_{min} was determined according to VDA 5, for which the MS is capable (see the bottom of Table 4).

Table 4. AACMM measurement results according to ISO 22514-7:2021.

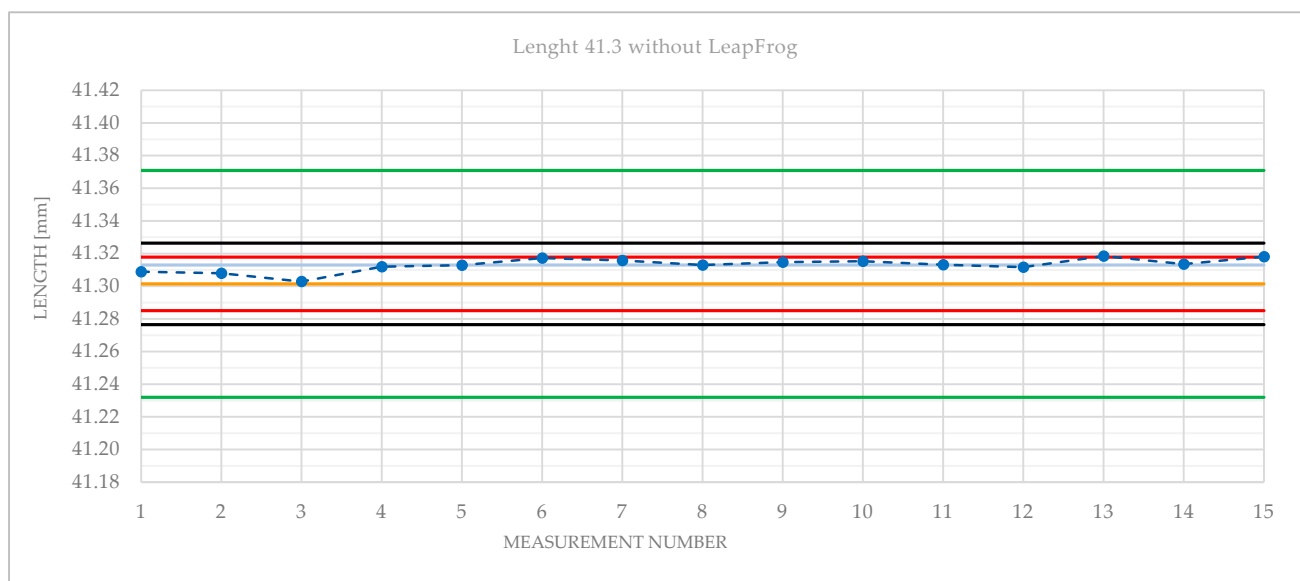
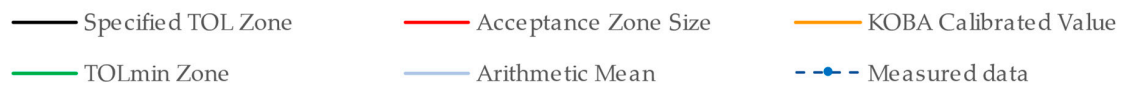
ISO 22514-7:2021	Requirements	Without Leapfrog	With Leapfrog
<i>TOL</i> (mm)	0.05	-	-
% <i>RE</i> (%)	≤5% <i>TOL</i>	2	2
Q_{MS} (%)	≤15%	41.678	58.980
C_{MS} (-)	≥1.33	0.47987	0.33910
VDA 5 TOL_{MIN} (mm)	-	0.13893	0.19660

Table 5 presents the results according to ISO 14253-1:2017. Without using the leapfrog method, the acceptance zone is 0.0328 mm wide, which is 65.6156% of the specified tolerance. The AACMM is applicable for verifying the conformity with the specification for this tolerance, with a moderate requirement for production centering. Using the leapfrog method, the acceptance zone is only 0.0257 mm wide, which is 51.3412% of the tolerance. The AACMM is applicable for verifying the conformity with the specification for this tolerance but with a high requirement for production centering.

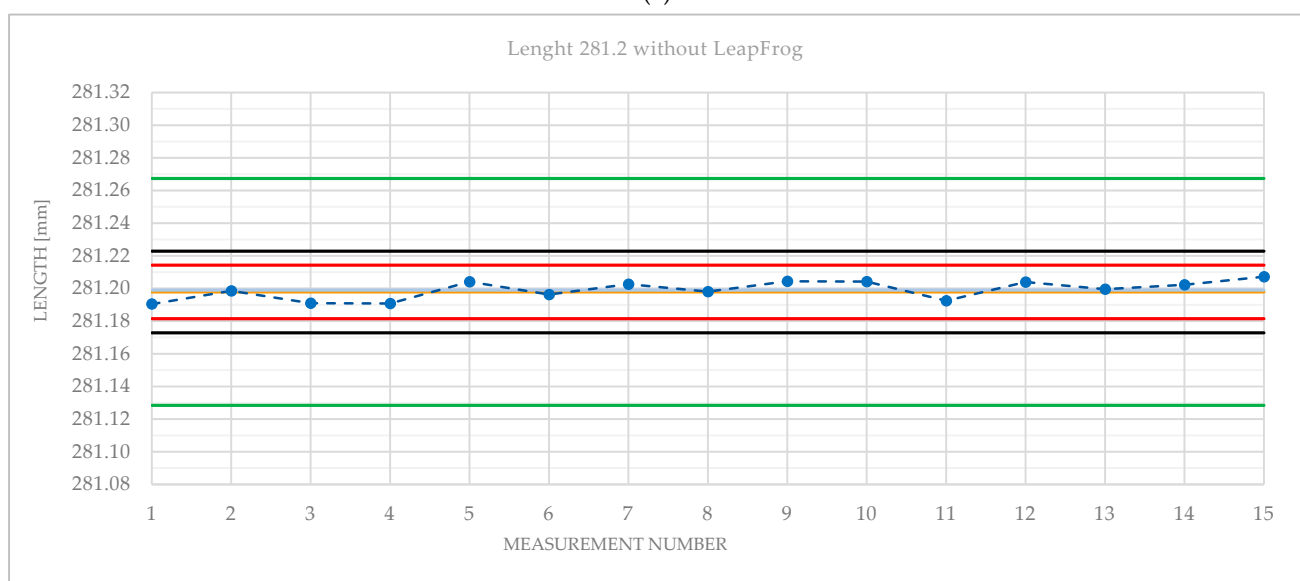
Table 5. AACMM measurement results according to ISO 14253-1:2017.

ISO 14253-1:2017	Without Leapfrog	With Leapfrog
Acceptance zone size (mm)	0.0328	0.0257
In % of tolerance	65.6	51.3

Figures 9 and 10 graphically present the measurement results obtained with the AACMM for all three measured lengths. Figure 9 shows the measurement without the leapfrog method, and Figure 10 shows the measurement with the leapfrog method. For each respective length, the size of the acceptance zone according to ISO 14253-1:2017, the size of the minimum tolerance TOL_{MIN} , and the position of the measured data are displayed.

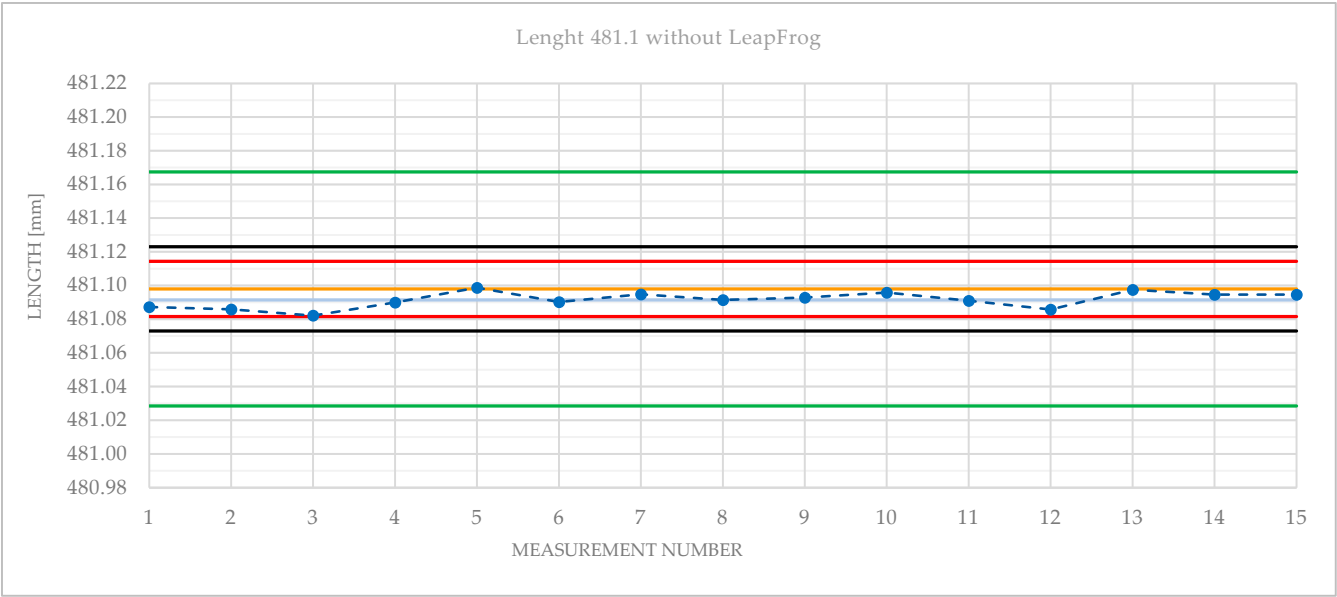


(a)



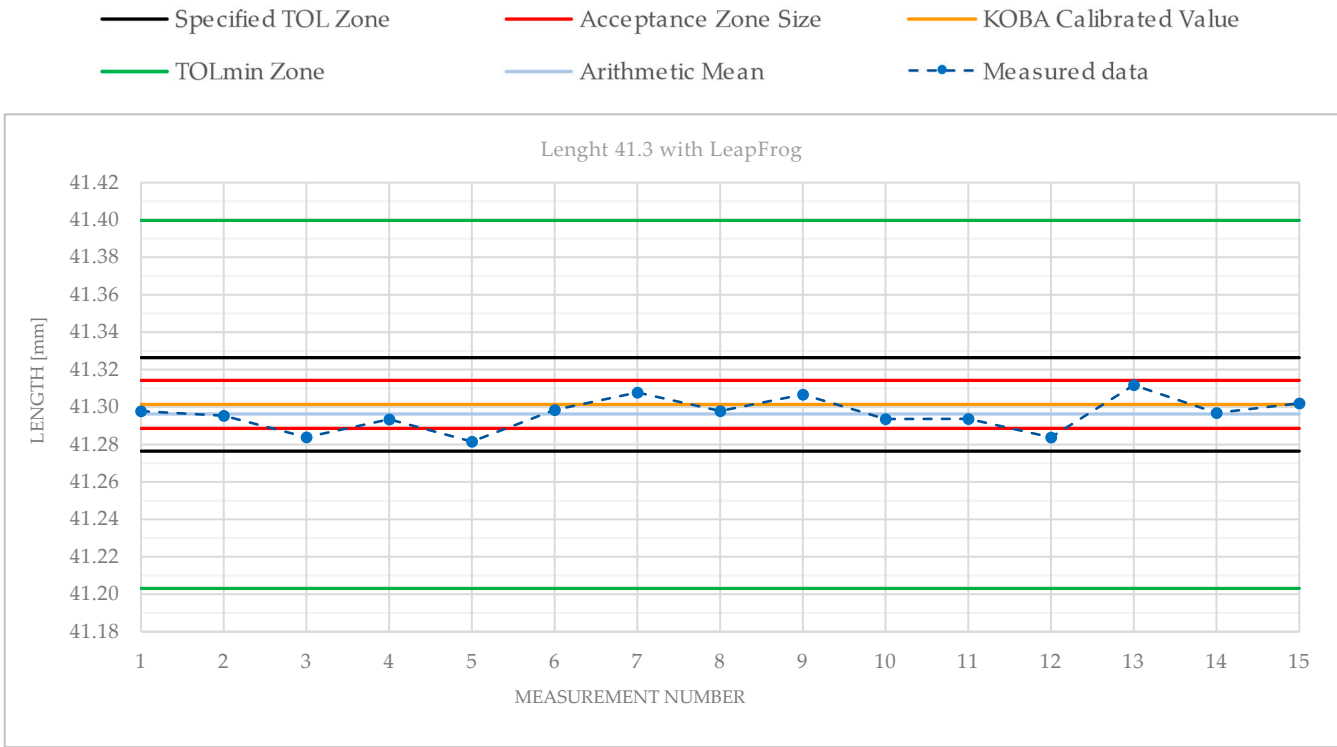
(b)

Figure 9. Cont.



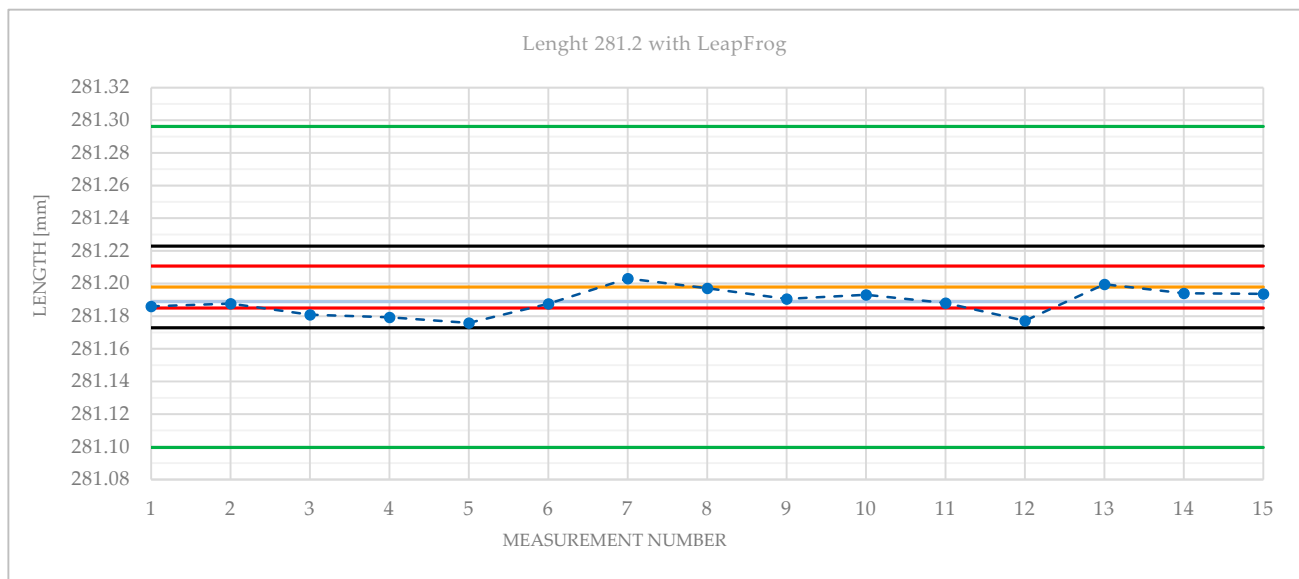
(c)

Figure 9. Graphical interpretation of measurement results without using leapfrog: (a) KOBA length 41.3; (b) KOBA length 281.2; (c) KOBA length 481.1.

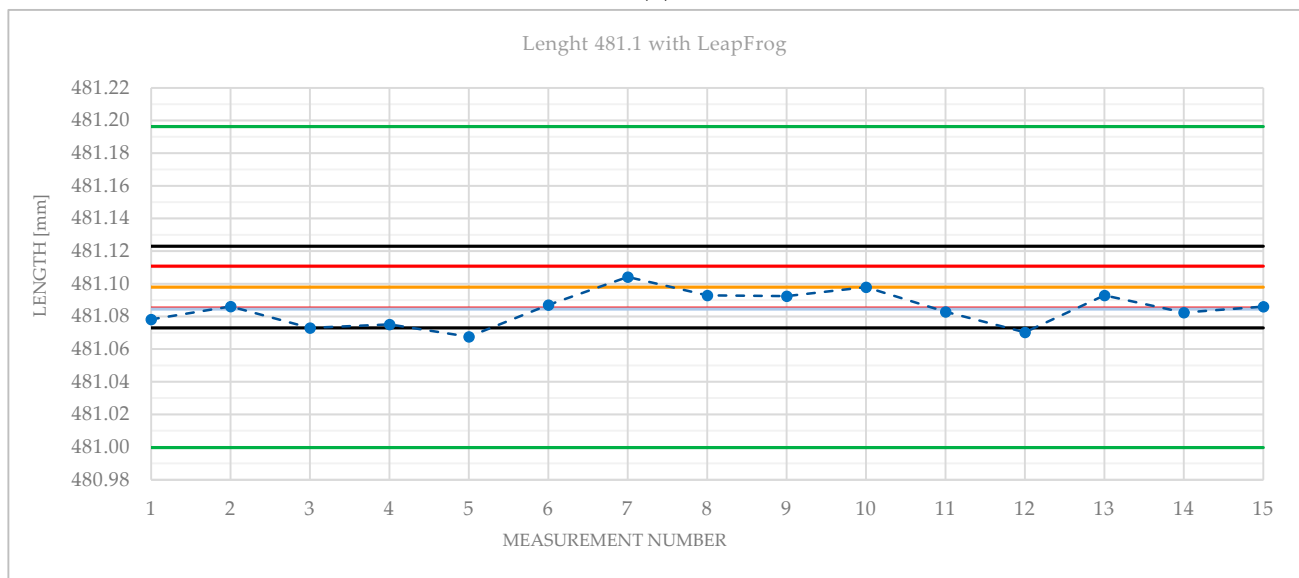


(a)

Figure 10. Cont.



(b)



(c)

Figure 10. Graphical interpretation of measurement results using: (a) KOBA length 41.3; (b) KOBA length 281.2; (c) KOBA length 481.1.

5.2. The Results of Measurements Using the Touch-Trigger Probe

Table 6 presents the set of measurement results obtained in accordance with ISO 22514-7:2021 for the RMP60 touch-trigger probe. Again, the MS has a satisfactory resolution. However, the values of the other two criteria do not meet the requirements in either case and therefore the MS is not capable of this tolerance value.

Table 6. Touch-trigger probe measurement results according to ISO 22514-7:2021.

ISO 22514-7:2021	Requirements	Touch-Trigger Probe Results
TOL (mm)	0.05	-
% RE (%)	$\leq 5\%$ TOL	2
Q_{MS} (%)	$\leq 15\%$	68.022
C_{MS} (-)	≥ 1.33	0.29402
VDA 5 TOL _{MIN} (mm)	-	0.22674

Table 7 presents the results according to ISO 14253-1:2017. The MS has a satisfactory resolution. For the specified tolerance of 0.05 mm, the acceptance zone is 0.0219 mm wide, which is 43.8821% of the tolerance. The device is suitable for verifying conformity with the 0.05 mm specification, with the highest requirement for production centering. Figure 11 graphically presents the measurement results using the RMP60 touch-trigger probe on the HCW 3 machine tool.

Table 7. Touch-trigger probe measurement results according to ISO 14253-1:2017.

ISO 14253-1:2017	Touch-Trigger Probe Results
Acceptance zone size (mm)	0.0219
In % of tolerance	43.9

— Specified TOL Zone

— Acceptance Zone Size

— KOBA Calibrated Value

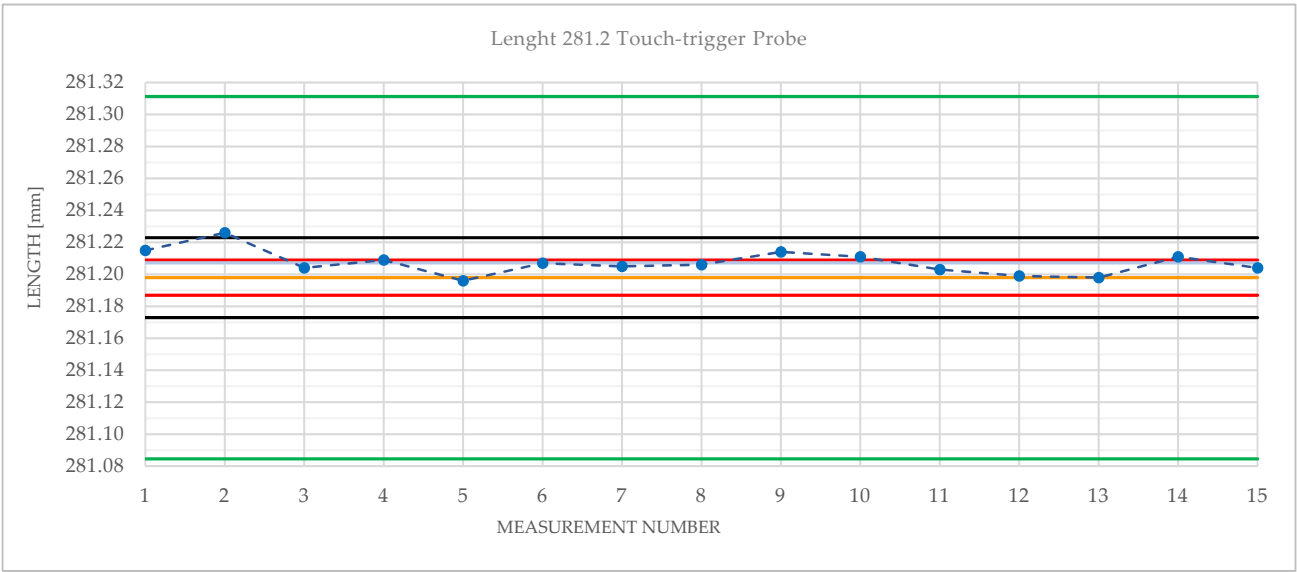
— TOLmin Zone

— Arithmetic Mean

— Measured data



(a)



(b)

Figure 11. Cont.

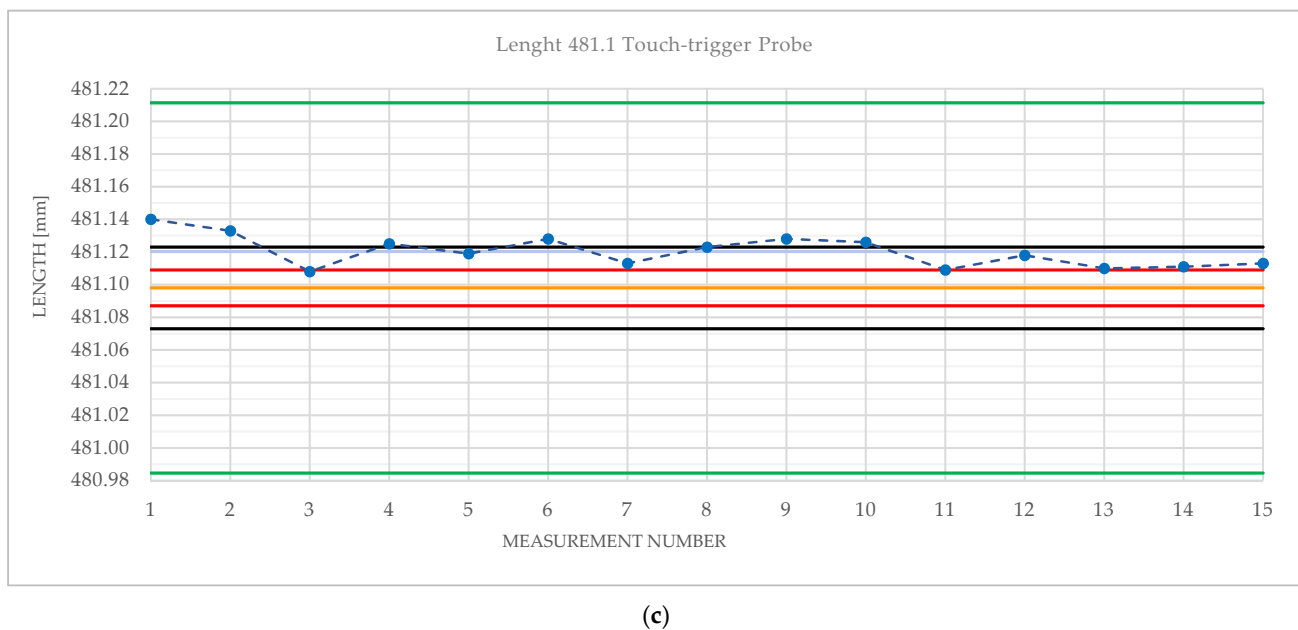


Figure 11. Graphical interpretation of measurement results with touch-trigger probe: (a) KOBA Length 41.3; (b) KOBA Length 281.2; (c) KOBA Length 481.1.

6. Discussion

In the presented case study, the procedures for evaluating MS capability according to ISO 22514-7:2021 were applied to two MSs: a TTP on a machine tool and an AACMM. Since these systems are inherently designed for measurements in a production environment, the testing was conducted under real conditions in an industrial setting. The results were compared with the possibilities of evaluating MS according to ISO 14253-1:2017, which does not directly assess the capability of the MS but rather whether the system is applicable for verifying the conformity with technical specifications. This evaluation method is applicable to a variety of measurement tasks with AACMM. Xie et al. [34] describe the integration of laser line scanners into AACMM. Experimental results showed a system error of approximately 0.2 mm on a test workpiece. Suresh and Dhanis [35] used AACMM to inspect holes after drilling with the scanning method and compared the results on a CMM. Scanning in combination with AACMM is significantly faster than CMM and at the same time has lower measurement uncertainty, which is also debatable given the relationships shown in Tables 54.1 and 54.2. Kaisatlis et al. [36] present the use of AACMM in combination with Industrial Total Station (ITS) in the field of Large-Scale Dimensional Metrology (LSDM). The aim was to develop a mathematical apparatus for the joint evaluation of inspected workpiece dimensions. In the case of such a deployment, the TOL_{min} of a measurement system composed of AACMM and ITS would be evaluated.

The results of this study revealed differences between the evaluation of MSs according to ISO 22514-7:2021 and ISO 14253-1:2017. When evaluated according to ISO 22514-7:2021, neither of the MSs (the TTP nor the AACMM) met the required Q_{MS} performance ratio criterion of $\leq 15\%$, nor did they meet the C_{MS} capability index criterion. This means that the expanded measurement uncertainty U_{MS} exceeded 15% of the total tolerance range. Under the given conditions, neither MS is sufficiently capable of measuring dimensions with a tolerance of 0.05 mm. The minimum tolerance TOL_{MIN} that both MSs can reliably measure was significantly higher than the specified value of 0.05 mm. Specifically, for the AACMM, this tolerance was calculated to be 0.13893 mm without leapfrog and 0.19660 mm with leapfrog, while for the TTP, it was 0.22674 mm. The interpretation of the results also highlights a significant difference between the minimum tolerance values according to ISO 22514-7:2021 and the acceptance zone size according to ISO 14253-1:2017. This difference

emphasizes how the more stringent capability requirements of ISO 22514-7:2021 lead to a higher minimum tolerance that the MS must meet to be considered capable.

When evaluated according to ISO 14253-1:2017, which allows us to determine the so-called applicability of MS for verifying the conformity with technical specifications on a drawing, both systems were conditionally deemed usable. ISO 14253-1:2017 does not directly assess MS capability but evaluates whether the MS can measure the specified tolerances which was achieved in this case. Although both systems exhibited higher uncertainty than would be acceptable for capability, their applicability to measure the specified tolerance of 0.05 mm, including the consideration of measurement uncertainties, could be confirmed. The degree of applicability, expressed as the ratio of the acceptance zone size to the considered tolerance for the tested MS, was evaluated according to the proposal from the article by Blecha et al. [8]. The conclusion of this article provides requirements for production centering, considering the ratio of the acceptance zone size to the considered tolerance as a percentage. Table 8 below presents these facts for the MS used in this case study.

Table 8. Production centering requirements according to ISO 14253-1:2017 results.

Measurement System	Acceptance Zone in % of TOL	Applicability/Capability	Requirements for Production
AACMM without Leapfrog	65.6	Conditionally applicable, Incapable	Increased requirements for production centering
AACMM with Leapfrog	51.3	Conditionally applicable, Incapable	Increased requirements for production centering
Touch-trigger Probe	43.9	Conditionally applicable, Incapable	Increased requirements for production centering

The presented results prompt reflection on why the tested MSs achieved rather poor results. Based on available sources and the results of this case study, it would have been surprising if the MSs had been deemed capable according to ISO 22514-7:2021. However, their applicability according to ISO 14253-1:2017 is also relatively low, and the pressure on production would be high.

Initially, the most surprising results come from the measurements with the RMP60 touch-trigger probe. The TTP installed on the HCW 3 machine tool performed the worst, and the minimum tolerance that the probe would be capable of is 0.23 mm. The graphs in Figure 11 show that the probe does not exhibit good repeatability. This raises the question of the condition of the specific HCW 3 machine, as it is fundamentally crucial when using a CNC machine tool as a measuring device to ensure that the results obtained are reliable. Of course, another key point of this study is that the experiments were conducted directly in a production environment, not under laboratory conditions. This introduced the influence of real-world conditions, such as vibrations, temperature changes, and other factors that could affect measurement stability. Measuring in a production environment places higher demands on the robustness and reliability of MSs and raises the question of whether it is possible to achieve the required accuracy under such conditions. The ambient temperature in the vicinity of the HCW 3 machine was monitored, and its fluctuation within 1 °C was deemed acceptable. It is important to note that the temperature conditions during the test were rather exceptional, and in other seasons, significantly greater temperature fluctuations, which have a substantial impact on machining and measurement, can be observed in this industrial facility [7]. Therefore, in this particular measurement, temperature may not be the most significant factor. Based on the measured data, the influence of certain types of vibrations or the geometry of the machine tool appears to be more likely.

At this point, it is worth noting that another MS suitable for in-process control, particularly of oversized parts mounted on machine tools, is the laser tracker. This is a portable measuring system that offers high accuracy and a large measurement range. Additionally, it can be positioned so that it is not directly mounted on the machine itself, creating an independent MS that maintains high measurement accuracy even over long distances.

However, this MS is also designed for use in a production environment and is therefore influenced by metrological conditions. At the time of writing this article, the authors could not find publications evaluating the capability of laser trackers. This could be a focus for future research. Such a measurement method could eliminate the influence of the condition of the machine tool. When measuring oversized parts with AACMM, it is easy to reach the limits of the measurement range, and leapfrog must be used. The laser tracker can also be relocated in a similar manner, but the system's measurement volume is so large that, with an initial proper placement, further relocation of the laser tracker is often unnecessary [25,37,38].

Future work could focus on long-term testing and evaluation of the capability and applicability of MSs under different measurement conditions, such as:

- a. Measurement at different ambient temperatures and seasons;
- b. Different machine settings and geometric accuracy;
- c. Operator variation;
- d. Monitoring of vibration effects from internal machine sources (pumps, axis drives, etc.);
- e. Design of fixture for arm attachment and with integration of all necessary sensors.

7. Conclusions

In conclusion, it can be stated that, according to ISO 22514-7:2021, the tested MSs were not sufficiently capable of measuring the 0.05 mm tolerance. Therefore, it can also be claimed that these in-process measurement methods are not comparable to laboratory measurements on CMMs in this respect. From the perspective of time savings and reducing the costs associated with the logistics of transporting workpieces to the laboratory, the efficiency of the measurement would be higher.

A number of studies focusing on the capability of MS, particularly on machine tools, have generally reached similar conclusions. It can be broadly inferred that all tested MSs designed for measurement in a production environment were applicable to measure specified tolerances but were not capable according to ISO 22514-7:2021 or VDA 5. It is important to explain that even though an MS does not meet the capability requirements of ISO 22514-7:2021, it can still be functional and applicable according to ISO 14253-1:2017. This means that in industrial practice, MSs can continue to be used for less stringent applications, where the primary goal is verifying conformity with technical specifications. If MS capability were to be considered as the sole criterion, the applicability of systems that have been successfully deployed up to now would be called into question. It would be simplistic to focus only on the capability of the MS when selecting it, without considering other practical factors. In the environment of industrial companies that machine oversized parts, logistics for transportation is a major consideration. Therefore, it is always necessary to consider whether it is appropriate to impose disproportionately high capability requirements on manufacturers' measuring equipment, even though, under other conditions, they are able to manufacture and measure parts in a manner that is applicable for the given tolerance.

The present research shows that both the AACMM and the TTP can be successfully integrated into in-process control if they are properly calibrated, used for applications where the tolerances match their capabilities, and if the machine tools are in good technical condition, particularly in terms of geometric accuracy and thermal stability. This ensures effective quality control without unnecessary logistical costs.

In summary, the following conclusions can be drawn:

1. Measurement with the AACMM without leapfrog has the biggest acceptance zone in % of TOL, approx. 65.6% followed by the AACMM with leapfrog (51.3). The touch-trigger Probe has an acceptance zone of approx. 43.9% of the tolerance.
2. The minimum tolerance TOL_{MIN} that both MSs can reliably measure was significantly higher than the specified value of 0.05 mm. Specifically, for the AACMM, this tolerance was calculated to be 0.13893 mm without leapfrog and 0.19660 mm with leapfrog, while for the TTP, it was 0.22674 mm

3. The interpretation of the results highlights a significant difference between the minimum tolerance values according to ISO 22514-7:2021 and the acceptance zone size according to ISO 14253-1:2017. This difference emphasizes how the more stringent capability requirements of ISO 22514-7:2021 lead to a higher minimum tolerance that the MS must meet to be considered capable
4. ISO 14253-1:2017 does not directly assess the MS capability but evaluates whether the MS can measure the specified tolerances, which was achieved in this case.
5. The known geometric accuracy of the specific machine is fundamentally crucial when using a CNC machine tool as a measuring device to ensure that the results obtained are reliable.

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Abbreviations

AACMM	Articulated arm coordinate measuring machine
ASME	American Society of Mechanical Engineers
C_g	Capability index
C_{gk}	Extended capability index
CMM	Coordinate measuring machine
C_{MS}	Measuring system capability index
CNC	Computer numerical control
C_p	Process capability
C_{pk}	Process capability index
g_a	Guarding band factor
GUM	Guide to the expression of the uncertainty of measurement
HCBW	Horizontal boring and milling machines
ISO	International Organization for Standardization
ITS	Industrial Total Station
k	Coverage factor
k_{CAL}	Coverage factor from the calibration certificate
L	Length
LSDM	Large-Scale Dimensional Metrology
MPE	Maximum permissible error
MS	Measuring system
MSA	Measurement System Analysis
MS_{EVR}	Pure Error Mean Square
MS_{LIN}	Lack of fit Mean Square
n	Number of measurements
OMM	On-machine measurement
Q_{MS}	Measuring system capability ratio
Q_{MS_max}	Maximum measuring system capability ratio
RE	Resolution

RMP	Radio machine probe
SMT	Skoda machine tools
T	Tolerances
TOL	Tolerance
TOL_{min}	Minimum tolerance
TPP	Touch trigger probe
U	Measurement uncertainty
u_{BI}	Standard uncertainty from the measurement bias
u_c	Combined measurement uncertainty
u_{cal}	Calibration standard uncertainty on a standard
u_{ever}	Standard uncertainty from repeatability on standard
u_{LIN}	Standard uncertainty from linearity of the measuring system
u_{MPE}	Standard uncertainty calculated based on maximum permissible error
u_{MS}	Combined standard uncertainty on measuring system
U_{MS}	Expanded uncertainty of the measuring system
u_{MS_REST}	Standard uncertainty from other influence components
u_{RE}	Standard uncertainty from resolution of measuring system
VDA	Verband der Automobilindustrie
$\bar{x}_{.321}$	Arithmetic mean of all the sample values
x_i	i th measurement input quantity
x_m	Reference quantity value

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